

AI Civic Issue Mapper – Complete Project Report

This report presents the complete plan and structure for the AI Civic Issue Mapper project. The system is designed as a government-style portal where citizens can report civic problems such as garbage dumping, broken roads, water leakage, or street light failures. The platform allows authorities to track, manage, and resolve issues efficiently while keeping citizens informed.

1. Project Objectives

- Provide a digital platform for citizens to report civic issues
- Help government departments track complaints efficiently
- Use AI to classify reported issues from images
- Improve transparency through complaint tracking

2. Key Features

- User registration and login (including Google login)
- Citizen dashboard
- Issue reporting with image and location
- AI-based issue classification
- Issue tracking system
- Admin dashboard
- Map visualization of complaints
- Notification system
- Analytics and reporting dashboard

3. System Workflow

- User registers or logs into the portal
- User accesses the citizen dashboard
- User submits a civic issue with description, image, and location
- System stores complaint and generates complaint ID
- AI model analyzes the uploaded image to detect issue type
- Admin views complaint from admin dashboard
- Admin assigns complaint to relevant department
- Department resolves the issue
- Admin updates complaint status

- Citizen receives status update and can track complaint

4. Modules of the System

- Authentication module
- Citizen dashboard module
- Issue reporting module
- Issue tracking module
- Admin management module
- AI classification module
- Map visualization module
- Notification module
- Analytics module

5. Database Components

- User table
- Complaint table
- Issue category table
- Admin table
- Notification table

6. Testing Strategy

- Frontend interface testing
- Backend API testing
- Database validation
- AI model accuracy testing
- Integration testing between modules
- Bug fixing and performance testing

7. Deployment Plan

- Prepare final tested version of application
- Deploy backend server
- Deploy frontend interface
- Connect domain or hosting platform
- Monitor application performance

Team Work Division (3 Members)

Role	Responsibilities
Frontend Developer	Design UI, build login/register pages, dashboard interface, issue reporting form, map
Backend Developer	Design database, implement authentication, store complaints, build admin dashboard
API / Testing / Integration	Develop APIs, integrate Google login API, connect frontend and backend, perform te

System Workflow Diagram (Text Representation)

- User Login/Register
- Citizen Dashboard
- Report Issue (Upload Image + Location)
- AI Issue Detection
- Save Complaint to Database
- Admin Dashboard Review
- Assign Department
- Update Status (Pending → In Progress → Resolved)
- Citizen Tracks Complaint
- Citizen Feedback